

Pairwise-nonorthogonal complex symmetric matrices without truncated Toeplitz models

Candidate counterexample packet — `agent_lane_04` (GPT5.6)

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Status. Candidate counterexample, likely valid; human review needed.

1 Source question

Garcia, Poore, and Ross ask in Section 7, Question 2 of [1] (PDF page 13):

Let $n \geq 4$. Is every complex symmetric matrix $M \in M_n(\mathbb{C})$ having no pair of orthogonal, nonzero eigenvectors unitarily equivalent to a truncated Toeplitz operator?

complex symmetric operators. For $n \geq 4$ numerical evidence strongly suggests that the implication (i) $\Leftrightarrow$ (ii) of Theorem 5.1 fails generically. In particular, (21) tends to fail for all $i = 1, 2, \dots, n$ for randomly generated complex symmetric matrices $M \in \mathbf{M}_n(\mathbb{C})$ which satisfy the hypotheses of Corollary 6.1. On the other hand, nothing that we know of prevents such a matrix from being unitarily equivalent to a truncated Toeplitz operator with symbol in L^∞ (as opposed to H^∞).

Question 2. Let $n \geq 4$. Is every complex symmetric matrix $M \in \mathbf{M}_n(\mathbb{C})$ having no pair of orthogonal, nonzero eigenvectors unitarily equivalent to a truncated Toeplitz operator?

A variant of the preceding is:

Question 3. Let $n \geq 4$. Is every irreducible complex symmetric matrix $M \in \mathbf{M}_n(\mathbb{C})$ unitarily equivalent to a truncated Toeplitz operator?

Recently, first author and J. Tener [16] showed that every complex symmetric matrix is unitarily equivalent to a direct sum of (i) irreducible complex symmetric matrices or (ii) matrices of the form $A \oplus A^t$ where A is irreducible and not UECSM (such matrices are necessarily 6×6 or larger). This immediately suggests the

Here “complex symmetric matrix” means $M^T = M$. Orthogonality in the question is for the usual Hermitian inner product on $\mathbb{C}^n$.

2 Result

Theorem 1. *For every $n \geq 10$, there exists $M \in M_n(\mathbb{C})$ such that*

- (i) $M^T = M$;
- (ii) M has simple spectrum and no two nonzero eigenvectors are orthogonal;
- (iii) M is not unitarily equivalent to a truncated Toeplitz operator.

Consequently Question 2 of [1] has a negative answer.

The theorem is existential. It proves that counterexamples occupy a nonempty part of a full-dimensional open semialgebraic locus after a lower-dimensional exceptional set is removed; it does not identify a particular rational or algebraic matrix outside that exceptional set.

3 Proof intuition

O’Loughlin associates to finite-dimensional truncated Toeplitz operators a semialgebraic set $\mathcal{U}(\mathcal{C}) \subseteq \mathcal{CS}_n$ and proves that its real dimension is strictly less than $n(n+1)$ for $n \geq 10$ [2]. It is not enough merely to choose a generic irreducible symmetric matrix: the source question imposes the sharper condition that every two eigenvectors be nonorthogonal.

The key observation is that this sharper condition is still generic on the simple-spectrum locus. It is semialgebraic, open, and nonempty. For nonemptiness, start with a complex orthogonal matrix $Q(t) = \exp(itK)$, where K is real skew-symmetric with no zero off-diagonal entries. Its columns are orthogonal for the bilinear form $x^\top y$, but, for small $t \neq 0$, every pair has nonzero Hermitian inner product because $Q(t)^*Q(t) = \exp(2itK) = I + 2itK + O(t^2)$. Giving those columns distinct eigenvalues produces the required symmetric matrices. A full-dimensional open semialgebraic set cannot be contained in O’Loughlin’s lower-dimensional TTO-model locus.

4 Proof

Let

$$\mathcal{CS}_n = \{M \in M_n(\mathbb{C}) : M^\top = M\},$$

viewed as a real vector space of dimension $n(n+1)$.

Define $\mathcal{G}_n \subseteq \mathcal{CS}_n$ to consist of matrices with simple spectrum for which eigenvectors belonging to distinct eigenvalues have nonzero Hermitian inner product. Because the spectrum is simple, this is equivalent to saying that no pair of nonzero eigenvectors is orthogonal.

Lemma 1. *For every $n \geq 2$, the set $\mathcal{G}_n$ is a nonempty open semialgebraic subset of $\mathcal{CS}_n$. In particular, $\dim_{\mathbb{R}} \mathcal{G}_n = n(n+1)$.*

Proof. The simple-spectrum locus is semialgebraic: it is defined by nonvanishing of the discriminant of the characteristic polynomial. Within that locus, the failure of pairwise nonorthogonality is the existence of $\lambda \neq \mu$ and unit vectors x, y satisfying

$$Mx = \lambda x, \quad My = \mu y, \quad x^*y = 0.$$

After real and imaginary parts are separated, these are polynomial equations and the inequality $|\lambda - \mu|^2 > 0$. Tarski–Seidenberg therefore shows that the bad set, and hence $\mathcal{G}_n$, is semialgebraic.

On the simple-spectrum locus the eigenlines vary continuously locally. The finitely many Hermitian inner products between normalized eigenvectors thus vary continuously up to harmless phase factors. Their nonvanishing is an open condition, so $\mathcal{G}_n$ is open in $\mathcal{CS}_n$.

It remains to show nonemptiness. Choose a real skew-symmetric matrix $K = (k_{ij})$ with $k_{ij} \neq 0$ whenever $i \neq j$; for example take $k_{ij} = 1$ for $i < j$ and $k_{ij} = -1$ for $i > j$. Put

$$Q(t) = \exp(itK).$$

Since $K^\top = -K$, we have $Q(t)^\top Q(t) = I$. Moreover itK is Hermitian, so

$$Q(t)^*Q(t) = Q(t)^2 = \exp(2itK) = I + 2itK + O(t^2).$$

For each $i \neq j$, its (i, j) entry has derivative $2ik_{ij} \neq 0$ at $t = 0$. Since there are only finitely many pairs, all off-diagonal entries are nonzero for every sufficiently small positive t .

Fix such a t and distinct complex numbers $\lambda_1, \dots, \lambda_n$. Set

$$M = Q(t) \operatorname{diag}(\lambda_1, \dots, \lambda_n) Q(t)^\top.$$

Then $M^\top = M$, and the columns q_j of $Q(t)$ satisfy $Mq_j = \lambda_j q_j$, since $Q(t)^\top Q(t) = I$. Their Hermitian Gram matrix is $Q(t)^* Q(t)$, whose off-diagonal entries are all nonzero. Hence $M \in \mathcal{G}_n$.

A nonempty open semialgebraic subset of the real vector space $\mathcal{CS}_n$ has the full dimension $n(n+1)$, proving the last assertion. $\square$

We now use the notation of O’Loughlin [2, Section 4]. Let $\mathcal{C} \subseteq \mathcal{CS}_n$ be the set of symmetric matrices N for which there is a diagonal matrix D with distinct unit-circle entries and $\operatorname{rank}[D, N] \leq 2$, and let $\mathcal{U}(\mathcal{C})$ be the set of symmetric matrices unitarily equivalent to some element of $\mathcal{C}$.

Theorem 3.1, Proposition 4.6, and the proof of Theorem 4.3 in [2] give

$$\dim_{\mathbb{R}} \mathcal{U}(\mathcal{C}) \leq 7n - 6 + \frac{n(n-1)}{2} < n(n+1), \quad n \geq 10. \quad (1)$$

The same paper’s Theorem 4.1 and Lemma 4.2 show that every finite-dimensional truncated Toeplitz operator has a symmetric matrix representative in $\mathcal{C}$. Consequently, any symmetric matrix unitarily equivalent to a truncated Toeplitz operator belongs to $\mathcal{U}(\mathcal{C})$.

For $n \geq 10$, Lemma 1 and (1) imply

$$\mathcal{G}_n \not\subseteq \mathcal{U}(\mathcal{C}),$$

because a semialgebraic subset cannot contain another semialgebraic set of strictly larger dimension. Choose $M \in \mathcal{G}_n \setminus \mathcal{U}(\mathcal{C})$. It is symmetric, has simple spectrum and pairwise nonorthogonal eigenvectors, and cannot be unitarily equivalent to a truncated Toeplitz operator. This proves Theorem 1.

5 Verification

Logical checks. The dimension comparison is strict already at $n = 10$:

$$7(10) - 6 + \frac{10 \cdot 9}{2} = 109 < 110 = 10 \cdot 11.$$

The difference grows thereafter. The TTO implication uses only the supporting paper’s stated matrix representation and rank-two commutator condition; no converse is assumed.

The semialgebraic description quantifies over normalized eigenvectors, so zero vectors cannot enter the projected bad set. Simple spectrum also handles two eigenvectors for the same eigenvalue: they are scalar multiples and cannot be orthogonal unless one is zero.

Computational sanity check. The included script constructs the displayed $Q(t)$ and M for $n = 10$, $t = 0.01$, and $\lambda_j = j$. Run from the repository root with

```
conda run --no-capture-output -n sandbox python \
  runs/fa_banach_001/solutions/counterexamples/\
  1012.4820_pairwise_nonorthogonal_tto_counterexamples/\
  code/verify_open_locus_example.py
```

It verifies complex orthogonality, symmetry, eigenvector residuals, nonvanishing Hermitian cross-inner-products, and the dimension arithmetic. The recorded output is:

```
n=10, t=0.01
max ||QTQ-I|| entry = 4.441e-16
max ||STT-S|| entry = 1.735e-17
min |<qi,qj>|, i<j = 0.020065386714
max eigenvector residual = 1.776e-15
TTO-locus dimension bound = 109
ambient/full-open-locus dimension = 110
strict dimension gap = 1
```

This computation proves neither the semialgebraic dimension theorem nor that this particular numerical matrix lies outside $\mathcal{U}(\mathcal{C})$; it is only a sanity check for Lemma 1.

6 Novelty check and limitations

On 26 August 2026, the four lightweight run indexes were searched for arXiv:1012.4820, the exact Question 2 wording, “pairwise nonorthogonal eigenvectors,” and truncated Toeplitz terminology. A bounded arXiv/web search used the exact phrases “no pair of orthogonal” and “pairwise nonorthogonal eigenvectors,” together with arXiv:1012.4820 and arXiv:2607.14019. It found the source paper and O’Loughlin’s later dimension paper, but no source stating the strengthened conclusion of Theorem 1. Novelty is therefore provisional rather than certified.

The argument is existential. It answers the universal question negatively and supplies counterexamples in every dimension $n \geq 10$, but it does not settle dimensions 4 through 9. It does not address Question 4 in [1], concerning matrices $A \oplus A^T$.

Human-review recommendation. Send to a reviewer familiar with semialgebraic dimension and truncated Toeplitz matrix models. The two main audit points are: (1) the projected semialgebraic description and openness of $\mathcal{G}_n$; and (2) the precise implication from a TTO model to membership in $\mathcal{U}(\mathcal{C})$.

References

- [1] S. R. Garcia, D. E. Poore, and W. T. Ross, *Unitary equivalence to a truncated Toeplitz operator: analytic symbols*, Proc. Amer. Math. Soc. 140 (2012), 1281–1295; arXiv:1012.4820.
- [2] R. O’Loughlin, *Semialgebraic Dimension and Truncated Toeplitz Models for Complex Symmetric Matrices*, arXiv:2607.14019, 2026.